Chapter 15
Development and Validation of the Perception of Interdisciplinary Research Collaboration (PIRC) Scale

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ABSTRACT

The perception of interdisciplinary research collaboration (PIRC) scale is a meticulously developed and validated tool comprising 64 items, crafted through concept analysis, content validity procedures, and expert panel refinement. Pilot testing involved 1,932 academic staff members from six universities in South-South Nigeria, revealing six dimensions: "challenges of IDR collaboration," "IDR collaborative experiences," "motivations for IDR collaboration," "benefits of IDR collaboration," "career impact of IDR collaboration," and "IDR team dynamics." Confirmatory factor analysis confirmed this structure, establishing strong convergent and discriminant validity. Reliability analysis indicated high internal consistency, making the PIRC scale a precise instrument for assessing interdisciplinary research collaboration. Its use enables stakeholders to understand researchers' perceptions, enhance collaboration, allocate resources efficiently, and foster innovation.

INTRODUCTION

Research is an action-oriented process involving a series of systematic procedures to create knowledge, identify problems and seek ways of dealing with such issues to improve man's understanding of the environment and other phenomena surrounding it. The primary essence of research is knowledge

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creation, modification and problem-solving (Odigwe et al., 2020; Owan & Bassey, 2019). This means that innovative approaches are required to keep the industry of knowledge creation and rapid evolution functional at all times for a better exploration and understanding of the environment (Bassey & Owan, 2018). Before the 1970s, research was bound by discipline (Allmendinger, 2015; Kulkarni, 2015; Van Rijnsoever & Hessels, 2011) where scholars of the same field partnered on research investigations for problem-solving and knowledge creation. In any case, as the focus started moving from fundamental investigations to settling greater difficulties (e.g., complex educational problems, environmental change, food and water emergency, general health, and recently, the emergence of big data and so on) research needed to rise above the limits of disciplines, making progress for interdisciplinary research.

Interdisciplinary research is a method of research by groups of people that coordinates information, data, strategies, apparatuses, viewpoints, ideas, hypotheses and theories from at least two disciplines to propel major understanding or to take care of issues whose solutions are beyond the extent of a single discipline. Klaassen (2018) offers an alternative definition, describing interdisciplinary research as the combination of methods, knowledge, skills, theories, and perspectives from different disciplines to foster innovative solutions and advance knowledge in uncharted problem areas. Interdisciplinary research varies from multidisciplinary research where specialists of different disciplines work independently on various parts of an expansive issue (Choi & Pak, 2007). Interdisciplinary research can likewise be distinguished from transdisciplinary research, where specialists absorb disciplinary-explicit theories and ideas to take care of an issue while limiting the isolation of the different disciplines (Fuqua, 2012).

This research approach has gained prominence, with institutional leaders championing interdisciplinary research, resulting in a rise in the number of interdisciplinary dissertations in recent years (Kniffin et al., 2021). Interdisciplinary research, characterized as the integration of knowledge, methods, and theories from various disciplines to tackle complex problems or gain a deeper understanding of a particular phenomenon (Bromham et al., 2016), has generated diverse findings in terms of its impact. Some studies suggest that interdisciplinary research can lead to greater benefits and influence, while others report mixed results or a lack of support for increased impact (Bromham et al., 2016). Since interdisciplinary research exceeds the regular scope of a discipline, numerous specialists and researchers across the globe accept that interdisciplinary research can resolve complex issues that solitary discipline research cannot (Barković, 2010; Kulkarni, 2015; Sherren et al., 2009). For instance, the field of computer science has been significantly influenced by interdisciplinary research, enabling the resolution of complex problems (Chakraborty, 2017). In addressing complex environmental issues, interdisciplinary research is often essential for developing a comprehensive understanding of integrated systems (Bark et al., 2016).

Therefore, the ascent of interdisciplinary research groups is expected, not to develop faith in heterogeneity, but rather to an expansion in the scientific unpredictability of modern issues requiring investigations. Interdisciplinary research has been gaining much attention in recent years and has been making more extensive social and economic contributions (Campbell et al., 2017; Feng & Kirkley, 2020; Rons, 2011; Urbanska et al., 2019). The underlying justification giving rise to IDR activities is traceable to an undeniably shared view among scholars and policymakers that some problems dwell beyond a single discipline and need multiple efforts, skills, techniques and expertise to handle. Another reason for IDR is the need to take care of the progressively complex issues of society (Allmendinger, 2015; Blackwell et al., 2010; Mainzer, 2011).

Various health, economic, educational, religious and social issues abound and keep recurring. Some of these issues are multifaceted and require a diverse approach to tackle them. The COVID-19 situation, which shuts virtually all sectors of the economies of almost every country of the world is an emerging instance of the complex situations of modern societies. Diverse efforts and collaboration are required to be able to handle the spread of the pandemic up to the development of a treatment or vaccine for curing infected persons. Apart from COVID-19, it is believed that the worst is yet to happen, and who knows, soon, worst situations may surface. This explains some of the reasons why the need for interdisciplinary research is widely advocated in domestic and international literature. Moreover, interdisciplinary research is increasingly being recognized and supported by funding agencies and institutions (Hirsch Hadorn et al., 2008). Many funding programmes now explicitly encourage interdisciplinary collaboration and provide dedicated funding streams for interdisciplinary research projects. In an instance, when scholars were solicited to submit cases from research that had a huge effect outside the scholarly community to the 2014 Research Excellence Framework (REF), 80% were discovered to be interdisciplinary (Kulkarni, 2015; Pedersen, 2016). Also, the Tertiary Education Trust Fund (TETFund) calls for the submission of grant proposals (from 2020 to date) and places serious emphasis and priority on transdisciplinary, multidisciplinary and interdisciplinary teams and projects. This explains the widespread attention that interdisciplinary research has gained in academia.

However, interdisciplinary research also poses challenges, such as the need for effective collaboration and the potential hindrance to the careers of early-stage researchers due to the pressure to publish in high-impact journals (Tappeiner et al., 2007). Nevertheless, while some scholars denounce the loss of disciplinary abilities and procedures, others have grasped the interdisciplinary transformation and its potential for interpreting scientific knowledge into cultural effect (Frodeman, 2011; Huutoniemi et al., 2010; Klein, 2010; Wagner et al., 2011). Supporters of interdisciplinary research point out that progress in interdisciplinary studies positively affects single-discipline studies (Kulkarni, 2015; Pedersen, 2016). In addition, interdisciplinary exploration gives more odds of making disclosures since combined information can prompt novel bits of knowledge. With research getting progressively worldwide and collective, interdisciplinary research is regarded to be the eventual fate of science and research.

Be that as it may, scholars who dive into interdisciplinary exploration face a few difficulties with accepting financing, receiving acknowledgement, and getting results published. It has also been discovered that attitudes toward interdisciplinary research vary, and the ongoing debate over the meaning of interdisciplinarity often centres around the concept of integration (Ahrens & Zascerinska, 2015). Top-level disciplinary journals are doubtful of distributing interdisciplinary studies, and there are scarcely any journals that distribute interdisciplinary studies solely (Kulkarni, 2015). In addition, many journal editors think that it is hard to choose reviewers who can assess interdisciplinary research. This affects the quality of review results as it is very difficult for a person to possess broad knowledge in all areas of research. This especially influences early career researchers, which is the reason they select to be a part of the ordinary single-discipline framework instead of digging into interdisciplinary exploration (Pannell et al., 2019).

Defenders of traditional disciplinary exploration opine that interdisciplinary research consumes funds and time from core disciplines. A potential reason for this could be the absence of satisfactory and fitting techniques for estimating the effect of interdisciplinary research. Researchers consent that tapping its effect is extreme since interdisciplinary research is mind-boggling, and the opinion is partitioned on whether it acquires more citations/references than disciplinary research. Moreover, quantitative measurements that are utilized to gauge the impact of disciplinary research do not support interdisciplinary exploration (Kulkarni, 2015). Another issue tied to IDRs is funding. Since the effect of interdisciplinary exploration is usually not prompt, funders may not see merit in putting resources into interdisciplinary research as is the case with single-discipline research.

Aside from issues identified with financing and publication, interdisciplinary research involves a more noteworthy challenge - the achievement of interdisciplinary tasks lies in different fields arriving at a shared belief (Kulkarni, 2015). The investigation of Buswell et al. (2017) recognizes a few reoccurring issues that are of significance in fruitful multidisciplinary coordinated efforts. These incorporate interpersonal relationships, time contributed and the space given to multidisciplinary working, area/location variation of the team members, regard for different disciplines, and trust between colleagues. Other key factors critical to powerful interdisciplinary research teams include "team purpose, goals, leadership, communication, cohesion, mutual respect and reflection" (Lakhani et al., 2012, p.260). This could potentially affect the perception and attitudes of scholars in interdisciplinary teams.

Nonetheless, the developing pattern of interdisciplinary research universally means that scholars of contrasting disciplines work together harmoniously and simultaneously. The difficulties confronting interdisciplinary research are many, however, significant partners of science, including funders, governments, and journals, have perceived the significance of supporting it (Bak et al., 2016). This study was undertaken given the perceived mixed perceptions surrounding the adoption of IDR as a model of research collaboration. Understanding the perceptions of scholars and researchers to collaborate in IDR teams may be useful in predicting future readiness to participate in IDR teams for the solving of complex problems facing man. Also, gaining an understanding of the perception of researchers towards IDR collaboration could inform us of the steps to take to innovate practices for the improvement of IDR success. Studying scholars' views on interdisciplinary research is also vital for improving collaboration, resource allocation, and educational programs. It informs strategies to overcome barriers, fosters innovation, and enhances problem-solving to address complex challenges.

In contrast, neglecting scholars' perspectives on interdisciplinary research has significant consequences. It can hinder the development of effective coping strategies for scholars working across disciplines, leading to difficulties in navigating complex interdisciplinary structures (Woiwode & Froese, 2020). This lack of insight impedes the provision of necessary support for successful interdisciplinary collaborations. Without studying scholars' perceptions, we may not fully grasp the potential benefits and drawbacks of interdisciplinary research or the factors that influence its success or failure, crucial knowledge for shaping policies and practices that enhance its impact. Furthermore, ignoring scholars' perceptions of interdisciplinary research can result in missed opportunities, inefficient resource allocation, ineffective collaboration, and limited progress in addressing complex challenges. It may also hinder educational programs and the development of evidence-based policies, potentially reinforcing inequalities in academia. The current study recognizes the significance of understanding researchers' perspectives on interdisciplinary collaborations. It aims to create and validate a tool for assessing how researchers perceive and approach engagement in interdisciplinary collaborations.

The need for an instrument to measure interdisciplinary research has been recognized by scholars and researchers (Nancarrow et al., 2013; Bosque-Pérez et al., 2016; Gould, 2015; Carr et al., 2018; Tate et al., 2018; Bridle et al., 2013; Brown et al., 2019; Begg et al., 2014; Handtke & Bögeholz, 2019; Meyers et al., 2012). Other scholars have highlighted the dearth or lack of instruments that can effectively assess and monitor the characteristics of interdisciplinary research teams (Porter et al., 2007; Rinia, 2007; Rons, 2011; Wagner et al., 2011). Furthermore, the systematic review conducted by a researcher indicated a need for an instrument to be developed for the assessment and monitoring of the characteristics of effective interdisciplinary research teams (Lakhani et al., 2012). To them, "such an instrument has the potential to provide information about interdisciplinary team attributes and to support team self-evaluation by identifying areas of strength and weakness" (p. E264). Interdisciplinary research teams play a crucial role in addressing complex problems and finding solutions (Bosque-Pérez et al., 2016). These teams consist of individuals with disciplinary depth, interdisciplinary breadth, and the ability to work with stakeholders to address problem-focused research questions (Bosque-Pérez et al., 2016). However, the effectiveness and productivity of interdisciplinary teams can be challenging to measure due to the lack of contextual detail and the complexity of interdisciplinary work (Nancarrow et al., 2013). Therefore, the development of an instrument specifically designed to measure factors contributing to interdisciplinary team effectiveness and productivity is essential (Nancarrow et al., 2013).

In addition to measuring team effectiveness and productivity, indicators of interdisciplinary research should also focus on collaborative processes within research teams (Tate et al., 2018). Interdisciplinary research is inherently collaborative, and the exchange of knowledge and engagement within teams is critical for its success (Tate et al., 2018). Understanding collaborative opportunities and barriers and assessing the readiness of teams to navigate them is crucial for effective interdisciplinary research (Tate et al., 2018). The development of an instrument to measure interdisciplinary research is particularly relevant in different fields (Gould, 2015). An instrument that can assess the characteristics and effectiveness of interdisciplinary research teams would be valuable in guiding the education and training of future researchers (Bosque-Pérez et al., 2016). To gain insight into interdisciplinary research and education programs, a framework for evaluation has been proposed (Carr et al., 2018). This framework can provide valuable guidance in assessing the impact and effectiveness of interdisciplinary programs, including research teams.

Some instruments have been constructed to measure factors contributing to interdisciplinary team effectiveness and level of productivity. Some of these instruments include the Myers-Briggs Type Indicator (MBTI) (Myers-Briggs, McCaulley & Most, 1985), Kolb Learning Style Inventory (LSI) (Kolb, 1984), Conflict Mode Instrument (CMI) (Kilmann & Thomas,1975) and the Bolman and Deal Leadership Orientation Instrument (LOI) (Bolman & Deal, 1991). The MBTI is a widely used instrument that assesses personality preferences and can provide insights into how individuals interact and communicate within a team (Nancarrow et al., 2013). The LSI measures an individual's preferred learning style, which can influence how they approach problem-solving and decision-making within a team (Körner, 2010). The CMI is a tool that assesses an individual's preferred approach to conflict resolution (Cassarino et al., 2018). Conflict is inevitable in interdisciplinary teams, and understanding how team members handle conflict can help facilitate productive discussions and prevent conflicts from escalating. The LOI measures an individual's leadership style and orientation (White et al., 2013). Effective leadership is crucial for interdisciplinary team effectiveness, and this instrument can help identify leadership strengths and areas for development within the team.

However, none of these measures have employed a theoretical framework to organize the concepts being assessed, nor have there been any psychometric evaluation of these measures (Lakhani et al., 2012). Therefore, a reliable and valid measure of the perception of staff towards engaging in interdisciplinary research collaboration would enable researchers and research teams to focus on specific areas of improvement, leading to greater productivity and maturity of their team. Such a tool would not only enhance the productivity and maturity of interdisciplinary teams but also optimize resource allocation, improve research outcomes, promote knowledge transfer, support interdisciplinary team development, and aid in academic and career advancement. It represents an investment in the future of science and innovation, facilitating effective collaboration to tackle some of society's most pressing issues.

PURPOSE OF THE STUDY

The purpose of this study was to develop and psychometrically validate and standardise an instrument that can be used to measure the perception of academics of interdisciplinary research collaboration. Specifically, the study:

- Established the content and face validity of the PIRC scale.
- Assessed the dimensionality and internal structure of the PIRC scale.
- Determine the Convergent and discriminant validity of the PIRC scale.
- Determine the reliability of the PIRC scale.

METHODS

Research Design

A cross-sectional survey research design was adopted by the researcher in developing and validating the PIRC scale. This design allows for the collection of data at one point in time from the participants. The design allowed the researcher to follow several steps in developing and validating the PIRC scale. These steps include concept analysis, content validity procedure, pretesting of items, pilot testing, extraction of factors, a test of dimensionality, test of reliability, test of validity, and the development of scoring and interpretation guidelines.

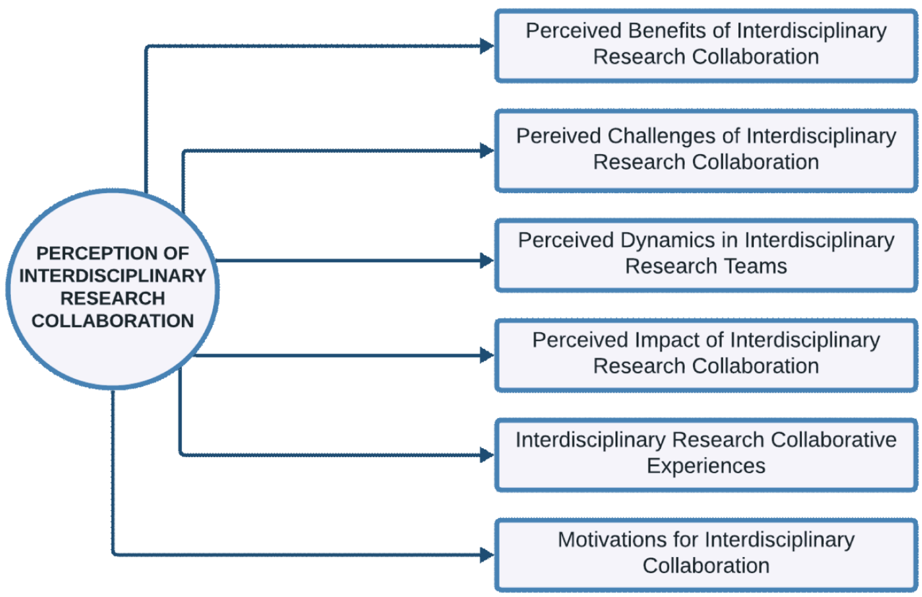
Concept of Interdisciplinary Research Collaboration

The concept of interdisciplinary research collaboration refers to the integration of knowledge and approaches from multiple disciplines to address complex problems (Petri, 2010). It involves the collaboration and cooperation of researchers from different disciplines, working together to achieve common goals (Feng & Kirkley, 2020). To measure the perception of interdisciplinary research collaboration, an instrument needs to be developed. The instrument should capture participants' perceptions of interdisciplinary collaboration, including their attitudes, beliefs, and experiences. It should assess factors such as the willingness to engage in interdisciplinary research, the importance of interdisciplinary work, and the perceived barriers and benefits of collaboration (Kirby et al., 2019).

Interdisciplinary research collaboration, in the context of this study, refers to the collaborative interaction and cooperation among individuals from diverse academic disciplines, backgrounds, and disciplines to tackle a research problem. It serves as a vehicle for addressing complex problems, fostering the generation of innovative solutions, and cultivating knowledge that transcends the confines of traditional disciplinary boundaries. Within this collaborative landscape, researchers engage in a dynamic exchange of ideas, methodologies, and insights, drawing from their varied perspectives to enhance the quality and impact of research outcomes.

The researcher developed a robust conceptual framework (see Figure 1) comprising several key dimensions based on information deduced from the literature. These dimensions encompass the benefits and advantages of interdisciplinary collaboration, the challenges researchers encounter in such endeavours, the dynamics within interdisciplinary teams, the career implications of engaging in interdisciplinary work, the actual experiences of researchers in collaborative projects, and the motivations that drive individuals to embark on interdisciplinary research collaborations. This framework provides a comprehensive lens through which researchers can gauge and evaluate the multifaceted nature of interdisciplinary collaboration and the perceptions associated with it.

Figure 1. Conceptual framework of perception of interdisciplinary research collaboration scale (Author's elaboration)



The Perception of Interdisciplinary Research Collaboration (PIRC) Scale

The PIRC scale is a self-report questionnaire developed to measure the perception of interdisciplinary research collaboration among researchers, academics, scholars and individuals engaging in research activities that transcend disciplinary boundaries. Through an extensive review of the literature, coupled with expert opinions from scholars with relevant experience, an initial pool of 80 items was developed. The items were worded in a manner that allows respondents to indicate (by ticking) the extent to which they agree or disagree with the items. The Likert scale ranged from 1 to 4, with the following response options: Strongly Disagree, Disagree, Agree and Strongly Agree.

Content Validity Procedure

To establish the content validity of the PIRC scale a panel of eight experts was formed. The experts were drawn from fields such as Educational Research, Measurement and Evaluation (n = 5), and Educational Psychology (n = 3). The ages of the experts ranged from 34 to 57 years, with work experience ranging from 9 to 22 years. The team of experts include three full professors and five associate professors. Regarding sex, five assessors were males and three were females. These experts, based on their diverse expertise, gender representation, age, extensive work experience, and academic ranks, independently reviewed and assessed the instrument items, based on criteria such as relevance, clarity, simplicity, and ambiguity. For each item, the experts assigned ratings, reflecting their varied perspectives and substantial experiences. Ratings were done on a scale of four across all the criteria. A total of 57 items that achieved a sufficiently high I-CVI of 0.78 or above on each of the four criteria were retained, following the recommendations of Lawshe (1975). However, seven items were revised, based on the remarks and suggestions of the experts since their I-CVIs ranged from 0.67 to 0.74. However, a total of 16 items were dropped for having very weak I-CVIs (below .50), reducing the number the length of the PIRC scale to 64 items.

Pretesting of Items

Before conducting the pilot testing, the instrument underwent a pretesting phase to assess its clarity, comprehensibility, and relevance. This was done through a focus group with a sample of potential respondents (Tripp & Shortlidge, 2020). The pretesting of instrument items involved 20 university lecturers with extensive interdisciplinary research experience recruited from two public universities in Nigeria. A purposive sampling strategy was adopted in recruiting the 20 participants to represent a diverse range of academic disciplines and backgrounds. The participants engaged in a focus group session guided by an experienced moderator and facilitator, both well-versed in interdisciplinary research dynamics. Informed consent was diligently obtained to ensure that there was voluntary participation. The researcher also assured them of confidentiality, promising them that all collected data would be anonymised.

During the structured focus group discussion, participants were presented with the revised version of the 64-item PIRC scale and encouraged to provide candid feedback. They highlighted aspects of the items that they found confusing, unclear, or irrelevant, and the facilitator used probing questions to elicit deeper insights. Qualitative feedback was meticulously audio-recorded and transcribed for thorough analysis. The resulting feedback drove item revisions to enhance clarity and relevance. This process ensured that the refined instrument effectively measured the perception of interdisciplinary research collaboration, setting the stage for subsequent validation and use.

Pilot Testing

For the pilot testing of the PIRC scale, the researcher selected a sample of academic staff with interdisciplinary research experience from six universities in South-South Nigeria. This choice ensured respondents possessed the necessary expertise to provide valuable feedback on the scale (Woosnam & Norman, 2009). To strike a balance between meaningful feedback and pilot study manageability (Teresi et al., 2021), power analysis guided the sample size determination (Cohen, 1988; Westland, 2010). An a priori sample size calculator for structural equation models (Soper, 2023) was used. The analysis assumed an effect size of 0.50 and a desired statistical power level of 80%. With six latent variables and 64 observed variables (reflecting the revised PIRC scale post-content validity and pretesting), a significance level of 0.05 alpha was set. The power analysis indicated a minimum required sample size of 40 participants to detect the effect size. However, the recommended minimum size based on the model structure was 1,989 participants.

Copies of the instrument were physically administered to a sample of 1,989 selected academic staff members. In conducting the pilot testing of the PIRC scale, ethical considerations were a top priority. The researcher took measures to protect participants' privacy and confidentiality, in line with ethical guidelines for research involving human subjects (Teijlingen et al., 2001). Informed consent was obtained from all selected academic staff members, who were fully informed about the study's purpose and the voluntary nature of their participation, with the right to withdraw at any point. Additionally, the researcher ensured that the survey was conducted onsite at each university. This arrangement guaranteed that respondents had access to the necessary resources and support to accurately complete the PIRC scale (Fappa et al., 2016). During the survey, participants were encouraged to provide feedback on each item of the scale through open-ended questions, allowing the researcher to identify any doubts or suggestions regarding the scale items and procedures. Following ethical protocols, the study received prior approval from the institutional review board of the University of Calabar. Upon completion of the data collection process, a total of 1,932 completed copies of the questionnaire were recovered and found useful, suggesting an attrition of 57 copies (and a rate of approximately 3%). Subsequently, the data collected from the pilot testing of the PIRC scale were analysed to assess the scale's psychometric properties.

RESULTS

Extraction of Factors

After the pilot testing, the data collected from the instrument were analyzed to identify the underlying factors or dimensions of interdisciplinary research collaboration. This was done using exploratory factor analysis, which will help determine the structure of the instrument and the relationships between its items (Ekpenyong et al., 2022, 2023; Owan, Owan, et al., 2022; Owan et al., 2023). The initial dataset, comprising responses to the 64-item instrument, underwent examination to ensure data quality. Checks for normality, outliers and multivariate outliers were conducted using histograms, boxplots, and Mahalanobis distance tests (Hair et al., 2019; Tabachnick & Fidell, 2019; Field, 2018; Stevens, 2012). Three multivariate outliers were detected by the Malahanobis distance test and deleted from the dataset, further reducing the number of cases from 1,932 to 1,929.

Principal Component Analysis (PCA) was chosen as the factor extraction method due to its versatility and applicability in exploratory factor analysis, which aligned perfectly to uncover latent dimensions without imposing a predefined factor structure. A combination of methods was employed to ascertain the number of factors to retain. The scree plot, displaying eigenvalues against the number of factors, revealed a clear inflexion point, indicating that retaining six factors was appropriate. Eigenvalues exceeding 1 supported this choice, and the conceptual interpretability of these six factors further solidified their selection. To enhance the interpretability of factors, a varimax rotation was applied. Varimax is an orthogonal rotation method that simplifies factor loadings by maximizing the variance of loadings on each factor (Kaiser, 1958). This rotation choice facilitated a clearer understanding of each dimension's content.

The analysis was performed initially, and 11 components were extracted which only explained 43.4% of the total variance extracted. However, a total of 25 items were screened for cross-loading unto multiple components, loading solitarily unto components, not loading unto a component in the matrix, and having loadings below the recommended threshold of .04 (Owan et al., 2021). After eliminating the 25 dysfunctional items, the analysis was re-run through PCA as the extraction method, with a varimax rotation to extract components based on the Eigenvalues above a value of one.

Six components were extracted, which jointly explained 72.19% of the extracted sums of squared loadings. The first component explained 14.65% of the extracted sums of squared loadings. Similarly, the second, third, fourth, fifth and sixth components explained 13.36%, 12.42%, 11.70%, 10.22% and 9.84% of the extracted sums of variance, respectively (See Table 1). The scree plot (Figure 2) also provides a visualisation of six factors having Eigenvalues above one. For sampling adequacy, a KMO value of .866 was obtained, with a significant Bartlett's test of sphericity, $\chi 2 = 68158.35$, $p < .001$. This shows that the sample of 1,929 respondents in the pilot sample is large enough and suitable for factor analysis.

Table 1. Results summary of the total variance explained by the extracted components in the PIRC scale

Components	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of S^2	Cum. %	Total	% of S^2	Cum. %	Total	% of S^2	Cum. %
1	5.71	14.65	14.65	5.71	14.65	14.65	5.17	13.26	13.26
2	5.21	13.36	28.01	5.21	13.36	28.01	5.05	12.95	26.20
3	4.84	12.42	40.43	4.84	12.42	40.43	4.79	12.28	38.48
4	4.56	11.70	52.13	4.56	11.70	52.13	4.65	11.93	50.41
5	3.98	10.22	62.34	3.98	10.22	62.34	4.28	10.98	61.39
6	3.84	9.84	72.19	3.84	9.84	72.19	4.21	10.80	72.19
7	0.95	2.44	74.62						
8	0.94	2.40	77.02						
9	0.85	2.17	79.19						
10	0.81	2.08	81.27						
11	0.63	1.62	82.89						
12	0.59	1.52	84.41						
13	0.50	1.28	85.69						
14	0.42	1.08	86.77						
15	0.40	1.02	87.79						
16	0.37	0.94	88.73						
17	0.35	0.90	89.63						
18	0.31	0.79	90.41						
19	0.30	0.77	91.18						
20	0.28	0.72	91.90						
21	0.26	0.67	92.57						
22	0.25	0.64	93.21						
23	0.24	0.61	93.82						
24	0.23	0.59	94.41						
25	0.22	0.55	94.96						
26	0.20	0.52	95.47						
27	0.19	0.50	95.97						
28	0.18	0.45	96.42						

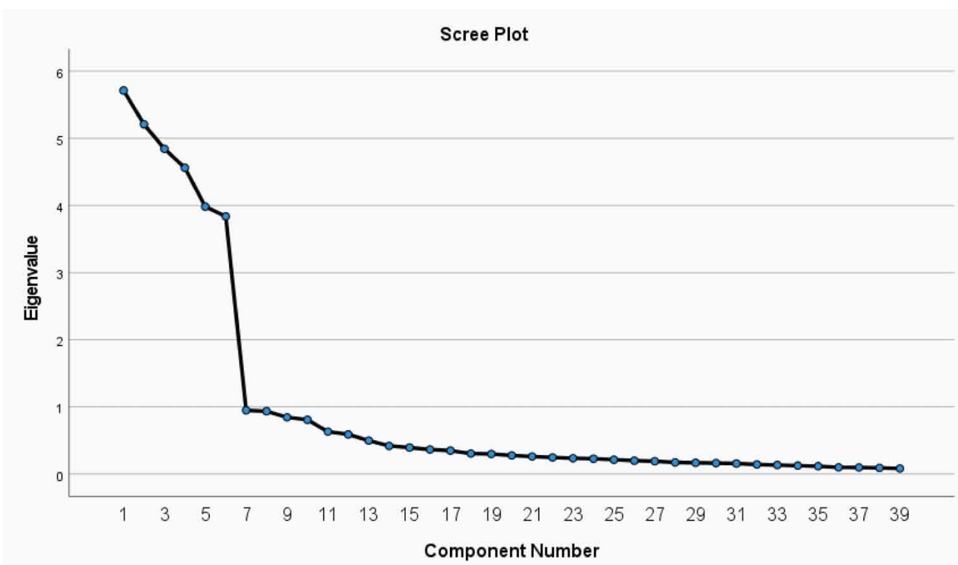
continues on following page

Table 1. Continued

Components	Initial Eigenvalues			Extraction Sums of Squared Loadings			Rotation Sums of Squared Loadings		
	Total	% of S^2	Cum. %	Total	% of S^2	Cum. %	Total	% of S^2	Cum. %
29	0.17	0.44	96.85						
30	0.16	0.42	97.27						
31	0.16	0.41	97.68						
32	0.14	0.37	98.05						
33	0.14	0.35	98.40						
34	0.13	0.33	98.72						
35	0.12	0.30	99.03						
36	0.10	0.26	99.29						
37	0.10	0.26	99.54						
38	0.09	0.24	99.78						
39	0.08	0.22	100.00						

Extraction Method: Principal Component Analysis.

Figure 2. Scree plot of the component structure of the PIRC scale



The rotated component matrix (Table 2) was evaluated for the specific loadings of each item to their corresponding components for interpretation of each of the six retained factors and naming purposes. The first component, denoted as "challenges of IDR collaboration," encapsulated items related to problems IDR teams face during collaboration. The item loadings in this component range from .726 to .896. The second component "IDR collaborative experiences," represented items reflecting the experiences of researchers engaged in interdisciplinary projects. Item loadings in this component range from .742

to .898. The third component, "motivations for IDR collaboration," included items probing into the motivating factors driving individuals to engage in interdisciplinary research collaboration. The item loadings within this component are in the range of .721 and .927.

The fourth component, "benefits of IDR collaboration," represents perceptions of the benefits and advantages of interdisciplinary approaches in the field of research. The items in this component loaded in the range of .665 and .935. The fifth component, "career impact of IDR collaboration," focused on items assessing the influence of interdisciplinary research collaboration on career development. The loading of the items in this component ranges from .616 to .836. The sixth component, "IDR team dynamics" represents positive dynamics and effective functioning of interdisciplinary teams in research. The item loadings within this component are in the range of .768 and .895.

Table 2. Rotated component matrix of the extraction solution for the PIRC scale

Item	Description	λ	Component
ITM1	An unequal number of members creates dominance of some disciplines over others in the team	.896	Challenges of IDR Collaboration
ITM7	Authorship disputes are common when publishing interdisciplinary research.	.886	
ITM8	Pursuing research agenda that aligns with the interests of all members can be challenging	.883	
ITM22	Misunderstandings in communication are common when collaborating across disciplines.	.883	
ITM21	Conflicting schedules make coordinating meetings with interdisciplinary team members	.881	
ITM31	The lack of experienced leaders affects the success rate of interdisciplinary research teams.	.826	
ITM14	Learning the language of another discipline is a barrier to interdisciplinary collaboration.	.726	
ITM5	Collaborative projects have generally improved the quality of my research.	.898	IDR Collaborative Experiences
ITM10	I have been satisfied with the outcomes of most of my collaborative research endeavours.	.892	
ITM23	My collaborative experiences have consistently led to the development of innovative ideas.	.884	
ITM12	I enjoy productivity when collaborating with colleagues from different disciplines.	.878	
ITM9	I have a sense of accomplishment from my past collaborative work.	.877	
ITM13	I have had the opportunity to learn other methodologies from interdisciplinary collaborators.	.748	
ITM3	Collaborating with colleagues from different disciplines has enhanced my skills.	.742	
ITM28	Breaking new ground in my research field motivates me to collaborate across disciplines	.927	Motivations for IDR Collaboration
ITM24	My goal is to expand my expertise when working with colleagues from other fields.	.925	
ITM32	Generating innovative solutions drives my motivation to collaborate outside my field.	.918	
ITM39	There is an experience of excitement while collaboratively tackling challenging problems	.915	
ITM34	I have a sense of accomplishment in addressing key issues through interdisciplinary research	.908	
ITM25	I strongly believe that interdisciplinary research always enhances my research quality.	.721	
ITM29	Integrating various research methods from different disciplines can tackle a complex problem	.935	Benefits of IDR Collaboration
ITM4	Collaborating with scholars from different fields increases exposure to diverse perspectives	.934	
ITM19	Interdisciplinary teams are more likely to receive funding for research projects.	.927	
ITM35	Some problems beyond the scope of a discipline warrant interdisciplinary collaboration.	.887	
ITM18	Interdisciplinary research can lead to discoveries that have a broader societal impact.	.883	
ITM15	Interdisciplinary collaboration fosters connections with scholars from various backgrounds.	.665	

continues on following page

Table 2. Continued

Item	Description	λ	Component
ITM11	Interdisciplinary projects have afforded me increased opportunities for self-development	.836	Career impact of IDR Collaboration
ITM26	My participation in interdisciplinary research projects has enhanced my visibility.	.831	
ITM16	Interdisciplinary collaboration is valuable for my career growth.	.829	
ITM33	The results of interdisciplinary collaborations have enhanced my research citations.	.826	
ITM30	Interdisciplinary collaboration has improved my prospects for securing research grants.	.813	
ITM17	Interdisciplinary collaborations have boosted my reputation beyond my field of research.	.670	
ITM2	Collaborating across disciplines has expanded my professional network.	.616	
ITM20	Interdisciplinary teams I work with often demonstrate a sense of unity.	.895	IDR team dynamics
ITM6	Disagreements are resolved constructively within our interdisciplinary teams.	.880	
ITM36	There are well-defined responsibilities that contribute to the smooth functioning of our teams.	.845	
ITM37	There is mutual respect among team members in my interdisciplinary research teams.	.812	
ITM27	Team members from different disciplines actively learn from one another.	.791	
ITM38	There is a sense of shared commitment to the success of our interdisciplinary projects.	.768	

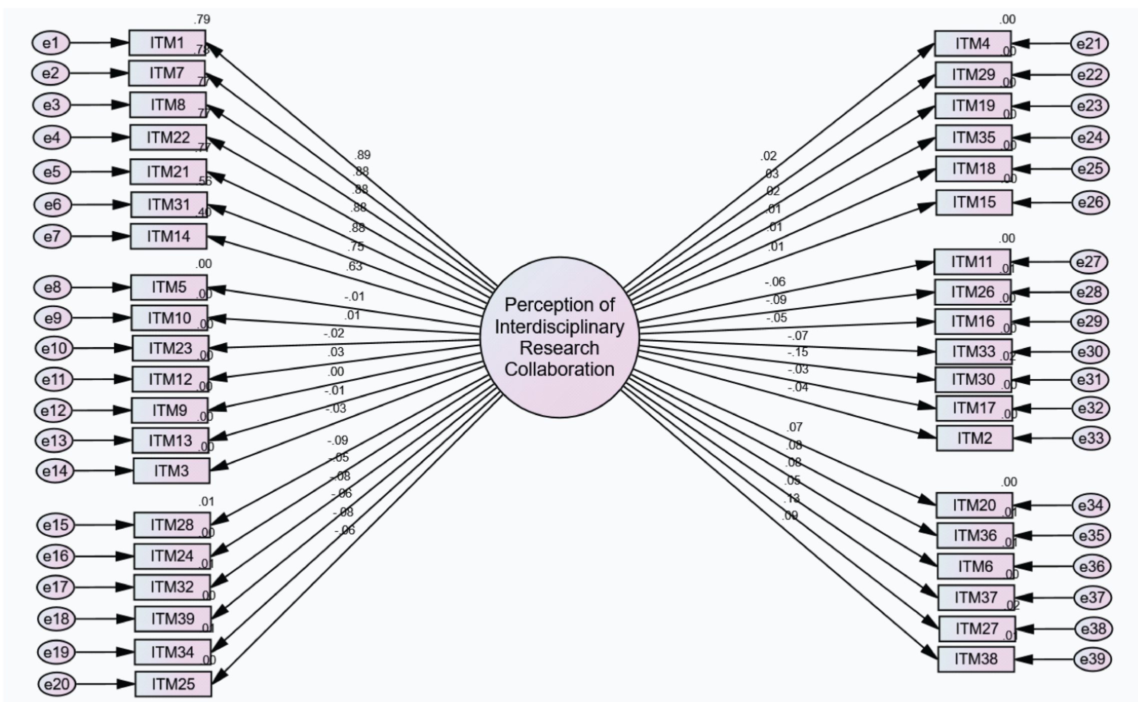
Notes: Extraction Method: Principal Component Analysis; Rotation Method: Varimax with Kaiser Normalization; Rotation converged in 5 iterations.

TEST OF DIMENSIONALITY

The test of dimensionality was conducted using confirmatory factor analysis to validate the factor structure identified in the previous step. This analysis assessed the fit of the data to the hypothesized factor structure and determined whether the instrument measures the intended dimensions of interdisciplinary research collaboration (Owan, Ekpenyong, et al., 2023; Owan, Odigwe, et al., 2022; Owan et al., 2021; Tripp & Shortlidge, 2020). In this study, four competing Confirmatory Factor Analysis (CFA) models were developed to further test the dimensionality of the PIRC scale. By comparing different models, the model that best captures the relationships between variables and provides the most accurate representation of the data can be identified (Booth & Hughes, 2014; Ganesh & Srivastava, 2022). Thus, this study aimed to identify the most suitable model that aligns with the complexity of the data. Below, the various competing models, including unidimensional, oblique, higher-order and bi-factor models are discussed.

The first model is the unidimensional or single-factor model (See Figure 3). This model posits that all instrument items measure a single underlying factor, representing a unidimensional perspective of interdisciplinary research collaboration perception (Rijnsoever & Hessels, 2011). The second is the oblique CFA model, used to examine the relationships between different dimensions or aspects of IDR collaboration (See Figure 4). Unlike the single-factor model, which assumes that all instrument items measure a single underlying factor, the oblique CFA model allows for the possibility of multiple underlying factors that are correlated with each other (Smith, 1996). The third is the higher-order CFA model (See Figure 5). The higher-order CFA model is used to assess the relationships between observed variables, latent factors, and higher-order latent factors (Shek & Yu, 2014). In this study, it was used to examine the relationships between different dimensions or aspects of collaboration, as well as the overall construct of interdisciplinary collaboration. The fourth is the bi-factor CFA model (See Figure 6). In the bi-factor model, the observed variables are grouped into specific factors that represent different dimensions or aspects of the construct being measured, and these specific factors are then grouped into a general factor that represents the overall construct (Rijnsoever & Hessels, 2011). In this study, the bi-factor model includes a general factor representing the overall perception of interdisciplinary research collaboration perception and six components, corresponding to distinct dimensions.

Figure 3. Unidimensional CFA model of the PIRC scale



In deciding the model that best fit the data, the four competing models were evaluated using seven fit indices, including the Chi-square (χ2), Standardised Root Mean Residual (SRMR), Tucker-Lewis Index (TLI), Comparative Fit Index (CFI), Root Mean Square Error of Approximation (RMSEA), Akaike's Information Criterion (AIC) and Bayesian Information Criterion (BIC). According to Table 3, all four models did not have an acceptable fit based on the Chi-square criteria since their p-values are less than the .05 alpha level. The result is understandable since the sample size of 1,929 is very large. The Chi-square test has been proven to be sensitive to sample size, with models exhibiting poor fits in samples larger than 400 participants (Bassey et al., 2019; Ekpenyong et al., 2022). Under the SRMR criteria, the unidimensional model exhibited the poorest fit of the four competing models. The oblique, higher-order and bi-factor models had acceptable fits based on SRMR criteria, with the oblique model having the best fit.

Figure 4. Oblique CFA model of the PIRC scale

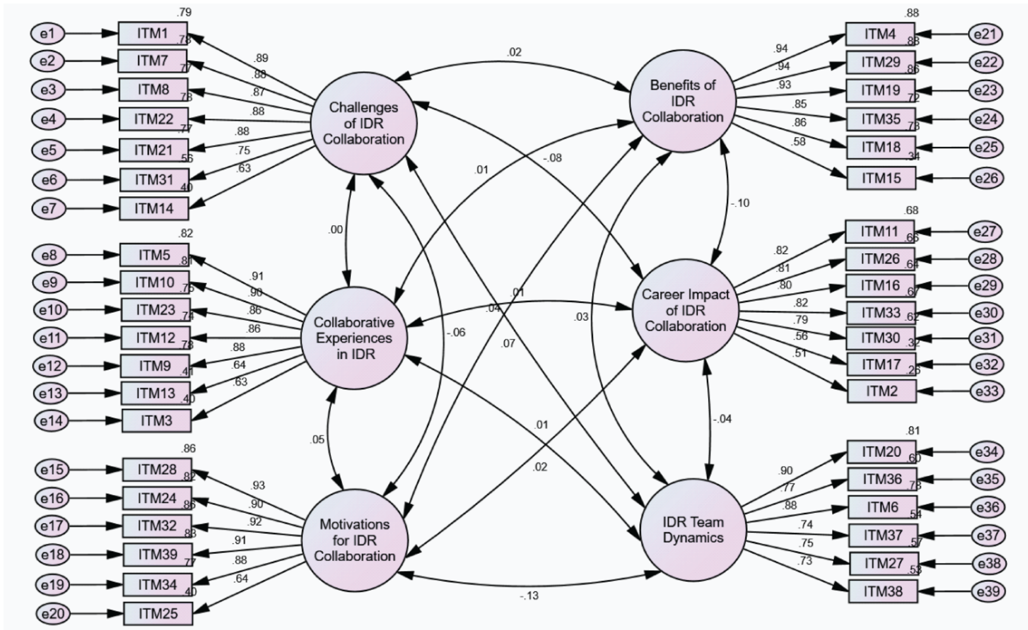


Figure 5. Higher-order CFA model of the PIRC scale

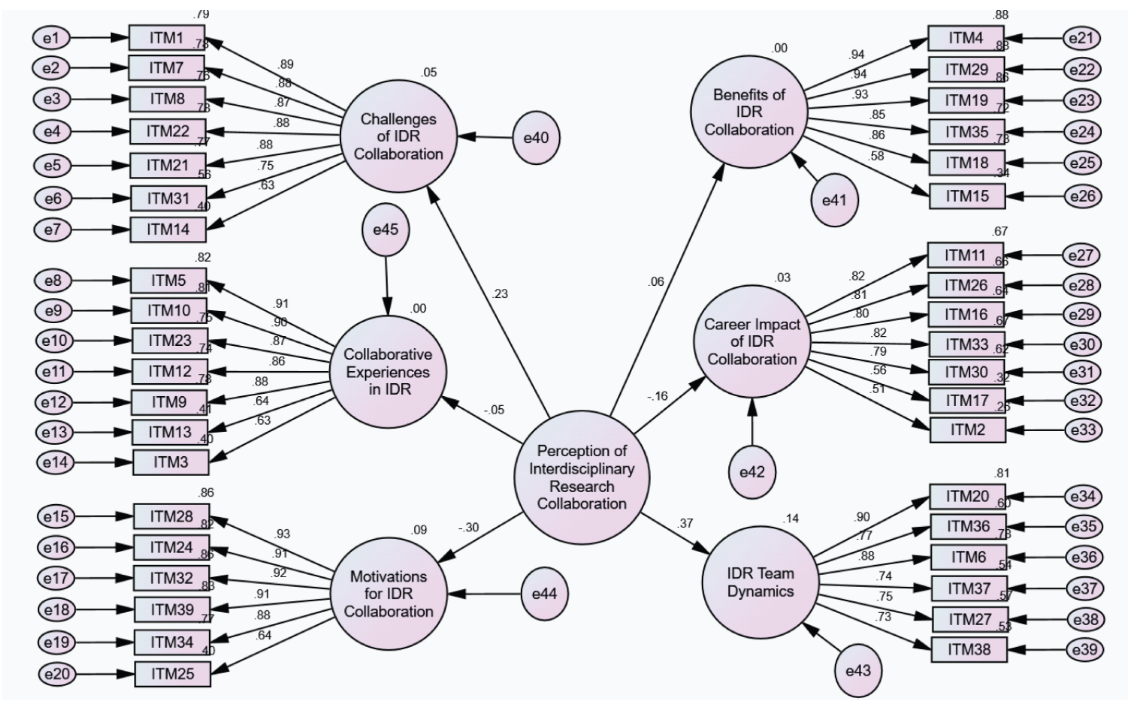
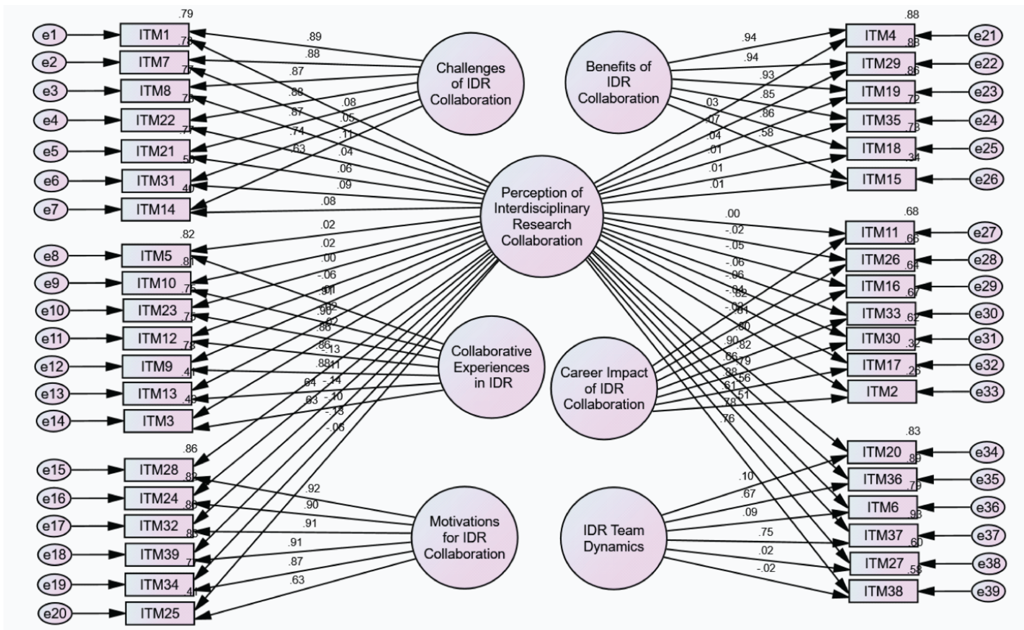


Figure 6. Bi-factor CFA model of the PIRC scale



For the TLI and CFI, all the models needed to attain values of .95 or higher for acceptability. However, none of the four models reached this threshold, although the oblique, higher-order and bi-factor models had TLI and CFI values all approaching the threshold. However, the unidimensional CFA model had TLI and CFI values closer to zero, making it the worst model in relation to the others. For the RMSEA, a value less than .80 is required and none of the models met this cut-off mark except the bi-factor model. The AIC and BIC also provided evidence that the bi-factor model has the best fit since compared to others, it has the lowest AIC and BIC. In conclusion, the bi-factor model provides the best fit, followed by the oblique model, and then the higher-order model. This suggests that there is a general factor (perception of interdisciplinary research) that influences the observed variables, as well as specific factors that capture unique variance in each variable. The Specific item loadings in the CFA models are provided in Table 4.

Table 3. Fit indices and model comparison of the four competing CFA models

Fit Index	Cut-Off	Model 1	Model 2	Model 3	Model 4
χ^2 (df), p-value	p > .05	57256.42(702), p < .001	9215.78 (687), p < .001	9246.77 (696), p< .001	7030.63 (663), p< .001
SRMR	< .08	.226	.039	.043	.042
TLI	.95	.121	.865	.866	.895
CFI	.95	.167	.874	.874	.906
RMSEA	< .08	.204	.080	.080	.071
AIC	NA	57412.42	9401.78	9414.77	7264.63
BIC	NA	57846.47	9919.30	9882.21	7915.71

Model 1 = Single factor model; Model 2: Oblique model; Model 3: Higher order model; Model 4: Bi-factor model;
SRMR = Standardised root mean residual; TLI = Tucker-Lewis Index; NA = Not applicable

Table 4. Loadings of confirmatory factor analysis for the four competing models

Paths for the General Factor	Model 1	Model 4	Paths for Specific Factors	Model 2	Model 3	Model 4
PIRC→ITM1	.89	.08	CIDR→ITM1	.89	.89	.89
PIRC→ITM7	.88	.05	CIDR→ITM7	.89	.89	.88
PIRC→ITM8	.88	.11	CIDR→ITM8	.88	.88	.87
PIRC→ITM22	.88	.04	CIDR→ITM22	.88	.88	.88
PIRC→ITM21	.88	.06	CIDR→ITM21	.88	.88	.88
PIRC→ITM31	.75	.09	CIDR→ITM31	.75	.75	.74
PIRC→ITM14	.63	.08	CIDR→ITM14	.63	.63	.63
PIRC→ITM5	-.02	.02	CEIDR→ITM5	.91	.91	.91
PIRC→ITM10	.01	.02	CEIDR→ITM10	.90	.90	.90
PIRC→ITM23	-.02	.00	CEIDR→ITM23	.87	.87	.87
PIRC→ITM12	.03	-.07	CEIDR→ITM12	.86	.86	.86
PIRC→ITM9	.00	.01	CEIDR→ITM9	.88	.88	.88
PIRC→ITM13	-.01	-.03	CEIDR→ITM13	.64	.64	.64
PIRC→ITM3	-.03	-.02	CEIDR→ITM3	.63	.63	.63
PIRC→ITM28	-.09	-.13	MIDR→ITM28	.93	.93	.92
PIRC→ITM24	-.05	-.11	MIDR→ITM24	.91	.91	.90
PIRC→ITM32	-.09	-.14	MIDR→ITM32	.93	.93	.91
PIRC→ITM39	-.06	-.10	MIDR→ITM39	.91	.91	.91
PIRC→ITM34	-.08	-.13	MIDR→ITM34	.88	.88	.87
PIRC→ITM25	-.06	-.06	MIDR→ITM25	.64	.64	.63
PIRC→ITM4	.02	.03	BIDR→ITM4	.94	.94	.94
PIRC→ITM29	.03	.07	BIDR→ITM29	.94	.94	.94
PIRC→ITM19	.02	.04	BIDR→ITM19	.93	.93	.93
PIRC→ITM35	.02	.01	BIDR→ITM35	.85	.85	.85
PIRC→ITM18	.01	.01	BIDR→ITM18	.86	.86	.86
PIRC→ITM15	.01	.01	BIDR→ITM15	.58	.58	.58
PIRC→ITM11	-.06	.00	CIIDR→ITM11	.82	.82	.82
PIRC→ITM26	-.09	-.02	CIIDR→ITM26	.81	.81	.81
PIRC→ITM16	-.05	-.05	CIIDR→ITM16	.80	.80	.80
PIRC→ITM33	-.07	-.06	CIIDR→ITM33	.82	.82	.82
PIRC→ITM30	-.15	-.06	CIIDR→ITM30	.79	.79	.79
PIRC→ITM17	-.03	-.04	CIIDR→ITM17	.56	.56	.56
PIRC→ITM2	-.04	-.02	CIIDR→ITM2	.51	.51	.51
PIRC→ITM20	.07	.90	TD→ITM20	.90	.90	.10
PIRC→ITM36	.08	.66	TD→ITM36	.77	.77	.67
PIRC→ITM6	.08	.88	TD→ITM6	.88	.88	.09
PIRC→ITM37	.05	.61	TD→ITM37	.74	.74	.75

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Table 4. Continued

Paths for the General Factor	Model 1	Model 4	Paths for Specific Factors	Model 2	Model 3	Model 4
PIRC→ITM27	.13	.78	TD→ITM27	.76	.76	.02
PIRC→ITM38	.09	.76	TD→ITM38	.73	.73	-.02

Model 1: Unidimensional or single-factor model; Model 2: Oblique or correlated factor model; Model 3: Higher order or hierarchical model; PIRC = Perception of interdisciplinary research collaboration; CIDR = Challenges of Interdisciplinary Research; CEIDR = Collaborative experiences of interdisciplinary research; MIDR = Motivations for interdisciplinary research; BIDR = Benefits of interdisciplinary research; CIIDR = Career impact of interdisciplinary research; TD = Team dynamics in interdisciplinary research.

CONVERGENT AND DISCRIMINANT VALIDITY

Performing convergent and discriminant validity assessments across various dimensions of the same instrument is crucial for establishing the reliability and validity of the instrument. Convergent validity refers to the correlation between the instrument and a theoretical dimension measure that targets the same aspects of the construct the instrument is designed to measure (Nunes et al., 2023). On the other hand, discriminant validity refers to the instrument's ability to distinguish groups in relation to the underlying construct (Kimberlin & Winterstein, 2008). The Fornell-Larcker approach to convergent and discriminant validity was employed in this study. The Fornell and Larcker approach is a widely used method for establishing convergent and discriminant validity in the context of covariance-based structural equation modelling (SEM) Henseler et al. (2014). This approach provides a systematic way to assess the extent to which a measurement instrument measures the intended construct and distinguishes it from other constructs.

Table 5. Convergent and discriminant validity evidence of the PIRC scale

Components	CR	AVE	1	2	3	4	5	6
1. Challenges of IDR Collaboration	.950	.733	**.856**					
2. IDR Collaborative Experiences	.947	.719	.000	**.848**				
3. Motivations for IDR collaboration	.957	.719	-.058	.058	**.848**			
4. Benefits of IDR Collaboration	.957	.790	.016	.013	.040	**.889**		
5. Career impact of IDR collaboration	.914	.769	-.075	.010	.016	-.085	**.877**	
6. IDR collaboration team dynamics	.931	.694	.079	.006	-.112	.025	-.040	**.833**

CR = Composite reliability – Values should be greater than .70
AVE = Average Variance Extracted – Values should be greater than .50
Bolded values along the leading diagonal are square roots of the AVE for discriminant validity.

To apply the Fornell and Larcker approach, researchers typically calculate the average variance extracted (AVE) for each latent factor (Owan, Emanghe, et al., 2022; Owan, Owan, et al., 2022). The AVE represents the amount of variance captured by the indicators of a latent factor relative to the measurement error (Fornell & Larcker, 1981). As shown in Table 5, all the components achieved convergent validity, since their Average Variance Extracted are greater than the recommended threshold of 0.50. To assess discriminant validity using the Fornell-Larcker criterion, the square roots of the AVE for each construct (bolded diagonal elements) were compared with the correlations between constructs (off-diagonal elements). As can be seen in Table 5, the square root of the AVE for each construct (the bolded values) in the leading diagonal are all greater than the correlation between each construct and other constructs.

TEST OF RELIABILITY

The reliability of the instrument was assessed to ensure that the PIRC scale consistently measures the perception of interdisciplinary research collaboration. This was done by calculating the degree of internal consistency using measures such as Cronbach's alpha (α), McDonald's omega (ω), and split-half reliability corrected with Spearman Brown's prophecy formula (rtt). The result of the analysis is presented in Table 6. Reliability estimates were computed for all the sub-scales and the overall instrument.

Table 6. Reliability estimates based on measures of internal consistency

S/N	Components	M	S^2	SD	k	ω	α	rtt
1	Challenges of IDR collaboration	17.61	42.25	6.5	7	.938	.939	.895
2	IDR collaborative experiences	24.93	104	10.2	7	.932	.934	.891
3	Motivations for IDR collaboration	18.18	54.92	7.41	6	.948	.946	.939
4	Benefits of IDR collaboration	21.56	78.86	8.88	6	.942	.938	.929
5	Career impact of IDR collaboration	17.24	35.79	5.98	7	.89	.891	.833
6	Team dynamics in IDR collaboration	17.34	49.28	7.02	6	.909	.913	.936
	Overall PIRC Scale	116.86	361.65	19.02	39	.668	.805	–

M = Mean; S2 = Variance; SD = Standard deviation; k = Number of items; ω = McDonald's omega; α = Cronbach's alpha; rtt = Split-half reliability coefficient corrected with Spearman Brown prophecy formula

According to Table 6, the reliability estimates for each sub-scale, across the three measures were above .70, indicating that items within each sub-scale are acceptable. The overall reliability of the PIRC scale is .805 for Cronbach's alpha and .668 for McDonald's omega. For Cronbach's alpha, this indicates that the items within the PIRC scale, when taken together, show a good level of internal consistency in measuring the same underlying construct. The McDonald's omega value is slightly lower than expected, indicating some variability in the internal consistency of the scale. This suggests that there may be some heterogeneity in the items, and the scale may not be as internally consistent as one would ideally want.

Furthermore, item-total correlation analyses were performed for each item within the instrument to examine the relationship between individual items and their respective factors (see Table 7). All items demonstrated robust positive correlations with their assigned factors, reinforcing the internal consistency of the instrument and the alignment of items with their intended dimensions. Specifically, the results in Table 7 showed that none of the items significantly affected the reliability of all the sub-scales if deleted. Consequently, no revisions were considered necessary to the items in the scale in this context.

Table 7. Assessment of the quality, impact, and reliability of individual items within each sub-scale

Components	Items	M	SD	Scale M if Item Deleted	Scale S^2 if Item Deleted	Corrected Item-Total r	Squared Multiple r	α if Item Deleted	ω if Item Deleted
Challenges of IDR Collaboration	ITM1	2.52	1.08	15.09	30.89	.851	.750	.925	.922
	ITM7	2.50	1.09	15.10	30.94	.832	.747	.927	.924
	ITM8	2.55	1.11	15.06	30.73	.835	.724	.927	.923
	ITM22	2.56	1.07	15.05	31.16	.830	.738	.927	.924
	ITM21	2.50	1.07	15.11	31.23	.828	.743	.927	.924
	ITM31	2.48	1.08	15.13	31.60	.777	.771	.932	.935
	ITM14	2.49	1.09	15.12	32.92	.654	.699	.943	.944
IDR Collaborative experiences	ITM5	3.54	1.68	21.39	76.38	.843	.790	.919	.914
	ITM10	3.59	1.69	21.34	76.37	.838	.777	.919	.915
	ITM23	3.58	1.76	21.35	75.46	.830	.716	.920	.915
	ITM12	3.68	1.74	21.25	76.02	.823	.715	.920	.916
	ITM9	3.57	1.73	21.36	76.30	.820	.745	.921	.916
	ITM13	3.48	1.71	21.46	80.21	.682	.695	.933	.936
	ITM3	3.49	1.74	21.44	80.00	.675	.688	.934	.937
Motivations for IDR Collaboration	ITM28	3.05	1.40	15.13	37.69	.889	.824	.930	.931
	ITM24	3.04	1.40	15.14	37.77	.887	.801	.930	.933
	ITM32	3.00	1.39	15.18	38.00	.878	.825	.931	.933
	ITM39	3.06	1.37	15.12	38.38	.867	.800	.933	.934
	ITM34	3.01	1.39	15.16	38.09	.871	.770	.932	.936
	ITM25	3.02	1.41	15.16	41.37	.636	.524	.959	.960
Benefits of IDR Collaboration	ITM4	3.68	1.67	17.89	54.14	.894	.843	.917	.922
	ITM29	3.58	1.69	17.98	53.67	.899	.846	.916	.921
	ITM19	3.60	1.71	17.97	53.77	.884	.824	.918	.923
	ITM35	3.58	1.67	17.99	55.41	.833	.704	.925	.931
	ITM18	3.51	1.75	18.06	54.62	.821	.712	.926	.931
	ITM15	3.62	1.68	17.94	61.02	.574	.365	.955	.956
Career impact of IDR collaboration	ITM11	2.48	1.10	14.76	26.11	.753	.640	.866	.864
	ITM26	2.47	1.09	14.77	26.25	.746	.605	.867	.864
	ITM16	2.47	1.08	14.77	26.36	.745	.574	.868	.865
	ITM33	2.45	1.13	14.79	26.00	.740	.638	.868	.865
	ITM30	2.49	1.11	14.75	26.28	.726	.584	.870	.867
	ITM17	2.43	1.11	14.81	27.84	.577	.434	.888	.892
	ITM2	2.46	1.08	14.78	28.57	.524	.408	.893	.896
Team dynamics in IDR collaboration	ITM20	2.85	1.41	14.49	33.62	.840	.742	.885	.876
	ITM36	2.93	1.41	14.40	34.51	.771	.841	.895	.898
	ITM6	2.89	1.41	14.45	33.86	.818	.715	.888	.880
	ITM37	2.93	1.42	14.41	35.12	.724	.823	.902	.904
	ITM27	2.81	1.39	14.52	35.62	.706	.548	.904	.898
	ITM38	2.93	1.37	14.41	36.21	.677	.528	.908	.902

M = Mean; SD = Standard deviation; S^2 = Variance; r = correlation; α = Cronbach alpha; ω = McDonald's omega

SCORING AND INTERPRETATION GUIDELINES

The PIRC instrument is designed to assess individuals' perceptions of interdisciplinary research collaboration. Each item in the instrument is scored on a four-point Likert scale, ranging from 1 (Strongly Disagree) to 4 (Strongly Agree). Participants' responses to individual items are summed to calculate their overall score. For each item, assign the following scores: Strongly Disagree: 1 point; Disagree: 2 points; Agree: 3 points; Strongly Agree: 4 points. Sum the scores for all items to obtain the participant's total score on the PIRC instrument. The total score for each dimension in the questionnaire will vary range depending on the number of items in the dimension. For the overall PIRC scale, a minimum score of 39 and a maximum score of 156 can be obtained. The interpretation of the PIRC scores is as follows:

Table 8. Scoring guidelines for the PIRC scale

S/N	Components	No. of Items	Score Range	Interpretation Cut-Off
1	Challenges of IDR collaboration	7	7 to 28	7 to 14 = low; 15 to 22 = moderate; Above 22 = high
2	IDR collaborative experiences	7	7 to 28	7 to 14 = low; 15 to 22 = moderate; Above 22 = high
3	Motivations for IDR collaboration	6	6 to 24	6 to 12 = low; 13 to 19 = moderate; Above 19 = high
4	Benefits of IDR collaboration	6	6 to 24	6 to 12 = low; 13 to 19 = moderate; Above 19 = high
5	Career impact of IDR collaboration	7	7 to 28	7 to 14 = low; 15 to 22 = moderate; Above 22 = high
6	Team dynamics in IDR collaboration	6	6 to 24	6 to 12 = low; 13 to 19 = moderate; Above 19 = high
	Overall PIRC Scale	39	39 to 156	39 to 78 = low; 79 to 118 = moderate; Above 118 = high; Above 118 = high

DISCUSSION

The present study aimed to develop and validate the Perception of Interdisciplinary Research Collaboration (PIRC) Scale, a self-report questionnaire designed to measure researchers' perceptions of interdisciplinary research collaboration. The study followed a rigorous methodology, including content validity procedures, pretesting of items, pilot testing, factor analysis, and assessments of dimensionality, reliability, and validity. The findings of this study have significant implications for understanding and assessing perceptions of interdisciplinary research collaboration among researchers and scholars. The factor analysis revealed a robust six-factor structure of the PIRC Scale, which accounted for 72.19% of the total variance. These factors represent distinct dimensions of researchers' perceptions related to interdisciplinary research collaboration. The identified dimensions include challenges of IDR collaboration, IDR collaborative experiences, motivations for IDR collaboration, benefits of IDR collaboration, career impact of IDR collaboration, and IDR team dynamics. This multifaceted structure underscores the complexity of interdisciplinary research collaboration and highlights the various factors that contribute to researchers' perceptions in this context.

The study employed confirmatory factor analysis (CFA) to validate the factor structure. Four competing models, including unidimensional, oblique, higher-order, and bi-factor models, were tested to determine the best-fitting model. This approach aligns with the one adopted by other studies. For instance, Hankins (2008) discusses the factor structure of the twelve-item General Health Questionnaire (GHQ-12) and compares competing models using confirmatory factor analysis. In the present the bi-factor model demonstrated the best fit, providing evidence that there is a general factor underlying perceptions of interdisciplinary research collaboration that influences individual dimensions. This supports the idea that there is a general factor underlying perceptions of interdisciplinary research collaboration that enables effective collaboration and knowledge integration (Lee et al., 2009).

One potential application of the bi-factor model is in the field of educational research. For example, a study by Chen et al. (2019) used the bi-factor model to examine the factor structure of a questionnaire measuring students' academic self-concept. The results indicated that a general factor of academic self-concept influenced specific factors such as math self-concept, reading self-concept, and science self-concept. In the field of psychology, the bi-factor model has been used to explore the factor structure of various constructs. For instance, a study by Reise et al. (2012) applied the bi-factor model to investigate the factor structure of the Inventory of Depression and Anxiety Symptoms (IDAS). The results revealed a general factor of emotional distress that influenced specific factors related to depression and anxiety symptoms. In the present study, the oblique model also showed acceptable fit indices, reinforcing the notion that the dimensions are related but distinct. These findings underscore the multidimensional nature of perceptions related to interdisciplinary research collaboration.

The study assessed convergent and discriminant validity using the Fornell-Larcker approach. The results showed that all components achieved convergent validity, with average variance extracted (AVE) values exceeding the recommended threshold of 0.50. Similar to the results of this study, some researchers also utilize the average variance extracted (AVE) and the Fornell-Larcker criterion to examine the discriminant validity of the scale (Salazar-Concha et al., 2022). Additionally, discriminant validity was supported, as the square roots of the AVE for each construct were greater than the correlations between constructs. These findings indicate that the PIRC Scale effectively measures the intended construct and can distinguish it from other related constructs.

Reliability analyses demonstrated the internal consistency of the PIRC Scale. Cronbach's alpha and McDonald's omega values for each sub-scale were consistently above 0.70, indicating good internal consistency. While the overall McDonald's omega value was slightly lower than expected, suggesting some variability in item consistency, it still indicated an acceptable level of reliability. Item-total correlation analyses further confirmed the internal consistency of the instrument, as no items significantly affected the reliability of the sub-scales when deleted.

Practical Implications

The development and validation of the PIRC Scale offer several practical implications. First, it provides researchers, academics, and scholars with a reliable and valid tool to assess perceptions of interdisciplinary research collaboration. This instrument can be valuable in both research and educational settings, enabling researchers to measure and understand the attitudes, beliefs, and experiences of individuals engaged in interdisciplinary research. Second, the identification of multiple dimensions within the PIRC Scale underscores the complexity of interdisciplinary research collaboration. Researchers and institutions can use this instrument to gain insights into the specific areas where improvements may be needed, whether in addressing challenges, enhancing collaborative experiences, or recognizing the benefits and career impacts of interdisciplinary work.

LIMITATIONS AND FUTURE DIRECTIONS

Despite its strengths, this study has some limitations. The sample used for validation was drawn from universities in South-South Nigeria, which may limit the generalizability of the findings to other cultural and geographical contexts. Future research should aim to validate the PIRC Scale in diverse settings to enhance its applicability. Additionally, while the PIRC Scale provides valuable insights into perceptions of interdisciplinary research collaboration, it does not capture the actual behaviours and outcomes of interdisciplinary research teams. Future studies could explore the relationship between perceptions and actual interdisciplinary collaboration outcomes to gain a more comprehensive understanding of the field. Therefore, the predictive validity of the PIRC scale should be evaluated using other outcomes such as research productivity, research impact, research funding acquisition, team formation, cross-disciplinary publications, innovation and creativity, and career advancement, among others.

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